

Question 1

Use the `gapminder` data set in the `gapminder` library to recreate a version of Hans Rosling's famous data visualization shown in the *A Grammar of Graphics* slides (a single plot instead of a movie; in other words, for just a single year).

- You can see at a glance which years are available by running `count(gapminder, year)` in an R chunk.
- In place of fertility rate use GDP per capita.

Constructing this plot requires several distinct steps.

1. Determine the correct geometry for the plot and make an initial `ggplot` with the two variables on the x and y axis.
2. Distinguish the continents by either shape or color. Which ever one you do not use, *set* its value to something other than the default. *Hint: use the help_file for `geom_point()` to find options you can set to!*
3. Alter the x and y axis labels so that they're more descriptive than just the variable given names in `gapminder`.
4. Add an annotation that draws attention to a particular feature of the data (of your choosing).
5. Title your plot with a claim based on your data.
6. Apply a theme of your choosing.

Use RStudio to write the code and see your visualization. Once you are happy with it, handwrite your code in the space below.

Question 2

Reconstruct a few of the visualizations shown in the *A Grammar of Graphics* textbook chapter using ggplot2 code. The datasets that you will need to access are penguins and arbutnnot; both are located in the stat20data package. Write the code you used below for each one, as well as the code that you used in any steps that you might need to take before making the visualizations.

- The scatter plot featuring penguin bill depths and bill lengths, broken down by species.

- The line plot featuring 17th century births in England.